

Name: _____

CIS 351 Sample CL3 Problem

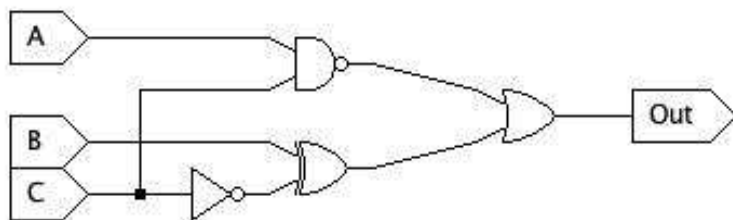
1 September 2026

CL3: Write Boolean Expressions

- (a) Write a Boolean expression in sum-of-product form that describes the following truth table:

x	y	z	
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

- (b) Write the Boolean expression that implements the following circuit:



CIS 351 Sample CL3 Problem Solutions

Thu 24th Sept, 2026

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$$\bar{x}\bar{y}z + \bar{x}y\bar{z} + xy\bar{z} + xyz$$

(b) Write the Boolean expression that implements the following circuit:

